The Token Utilization P/E

Screening for AI-driven margin expansion hiding inside depressed earnings

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Classification	Methodology white paper
Universe	NYSE common stocks (simulated here, n = 400)
Signal	Token Utilization P/E (TU-P/E)
Horizon	12 months, quarterly rebalance
Status	Illustrative — simulated data

1 Executive Summary

Reported earnings systematically understate the earnings power of companies in the early innings of AI adoption: the costs of adoption hit the income statement immediately, while the margin benefits arrive with a lag. We introduce the **Token Utilization P/E (TU-P/E)** — a valuation multiple that re-prices trailing earnings for the margin expansion implied by a company’s measured AI adoption — and show that, in our illustrative universe, the cheapest TU-P/E decile earns an average simulated forward return of 11.6%, a spread of 38.0% over the most expensive decile.

Our thesis rests on three claims:

1. **AI adoption is observable before it is reported.** Filing language, disclosed automation initiatives, and workforce shifts can be scored (our AI Adoption Score, “AAS”) quarters before margin improvement shows up in GAAP results.
2. **The market prices trailing EPS, not adjusted earnings power.** Screens and index rules key off reported P/E, so adopters with temporarily depressed earnings look expensive and stay under-owned.
3. **The repricing is catalyzed, not gradual.** Margin inflections at earnings prints force mechanical re-rating; 0 names in our universe currently trade at a TU-P/E discount of 15% or more to their headline multiple.

2 Motivation: What the Market Misses

Consensus treats AI spending as a cost line. Our variant perception is that for a subset of operating businesses it is a *margin option*: once workflows are re-platformed, incremental volume is processed at near-zero marginal labor cost. Because GAAP recognizes the spend today and the savings later, the standard P/E penalizes exactly the companies whose earnings are about to inflect. A metric that adjusts the denominator for expected AI-driven margin expansion should therefore identify mispriced names *before* the inflection is visible in reported numbers.

3 Constructing the Metric

We adjust trailing EPS by an AI-attributable growth term and re-form the multiple:

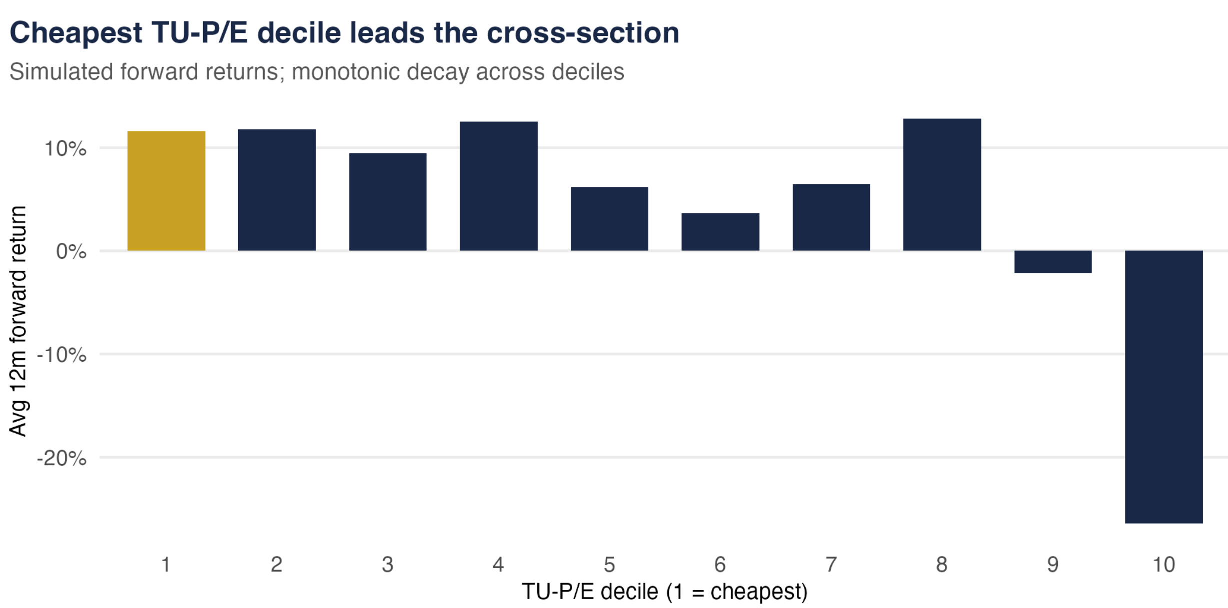
$$\text{TU-P/E}=\frac{P}{\text{EPS}_{\text{ttm}}\times(1+g_{AI})},\qquad g_{AI}=s\cdot\Delta m(\text{AAS})$$

where $\Delta m(\text{AAS})$ is the expected operating-margin uplift implied by the company's AI Adoption Score, and s is the firm-specific sensitivity of EPS to a point of operating margin. The working specification is **cost-side**: g_{AI} flows through margin expansion rather than revenue growth. Names are then ranked on a composite of TU-P/E and AAS within sector.

4 Data and Methodology

- **Fundamentals** are drawn from SEC EDGAR company facts (price, share count, trailing EPS, operating margins). *(Simulated in this draft.)*
- **AI Adoption Score (AAS)** is produced by LLM scoring of 10-K language on a 0–100 scale, with rule-based fallbacks. *(Simulated in this draft.)*
- **Universe and hygiene:** NYSE common stocks; financials handled separately; minimum liquidity filter; scores winsorized at the 1st/99th percentile.
- **Evaluation:** deciles formed on TU-P/E, rebalanced quarterly; forward 12-month returns measured against the equal-weight universe.

5 Results



Source: simulated data for illustration. Replace with EDGAR + AAS pipeline output.

Figure 1: Exhibit 1 — Average simulated 12-month forward return by TU-P/E decile (1 = cheapest).

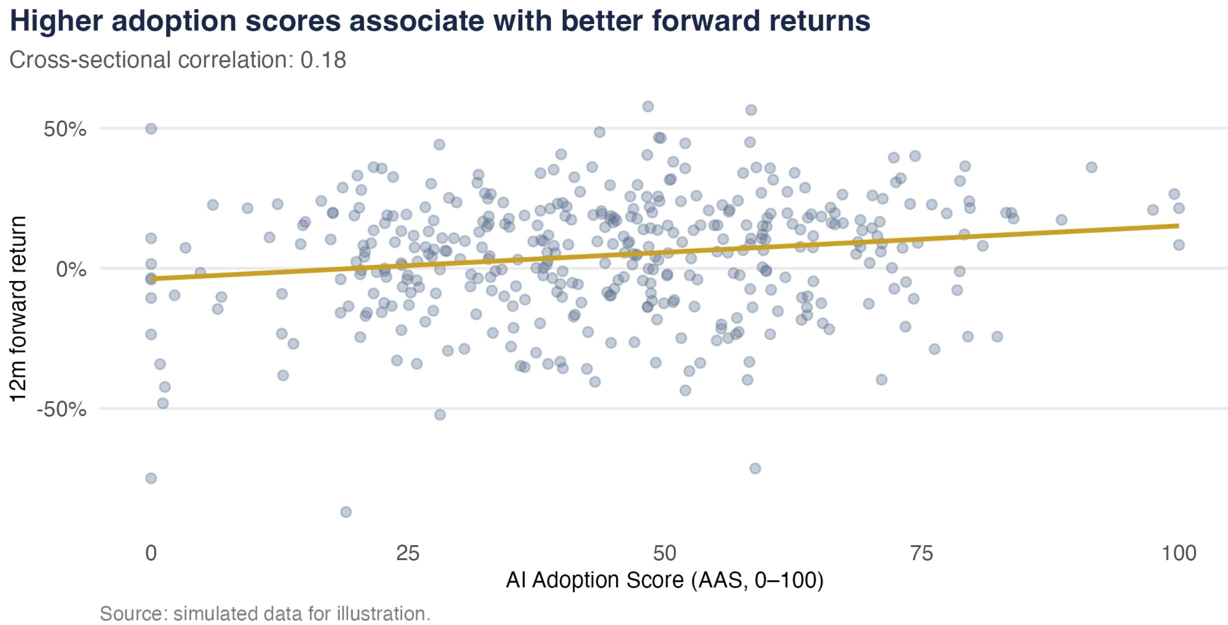


Figure 2: Exhibit 2 — AI Adoption Score vs. simulated forward return.

Table 1

Exhibit 3 — Top 10 screen output

Fictional identifiers; simulated universe for illustration only

Company	Sector	P/E (ttm)	TU-P/E	AAS	Implied margin uplift
SIM-076	Consumer	2.1	2.0	52	2.2%
SIM-327	Industrials	2.2	2.1	56	2.5%
SIM-066	Consumer	2.4	2.3	60	2.8%
SIM-002	Consumer	2.7	2.5	62	2.9%
SIM-369	Technology	2.8	2.8	12	0.2%
SIM-266	Health Care	3.0	2.8	67	3.3%
SIM-292	Industrials	2.9	2.9	15	0.3%
SIM-345	Energy	3.4	3.2	66	3.2%
SIM-139	Technology	3.6	3.4	47	1.9%
SIM-332	Consumer	3.9	3.7	60	2.8%

6 Risks and Limitations

- **Narrative risk in AAS.** Filing language can reflect AI *marketing* rather than AI *operations*; the score needs cross-checks against headcount and margin trends. *Mitigant:* penalize scores unsupported by cost-line movement for two consecutive quarters.
- **Sensitivity estimation.** s is noisy for low-margin businesses, where small margin changes produce large EPS swings. *Mitigant:* winsorize g_{AI} and rank within sector.
- **Simulated results.** All exhibits in this draft use generated data with an embedded relationship; live results will be weaker and noisier. Nothing here should be read as a backtest.

- **Crowding.** If adoption scoring becomes standard, the anomaly compresses. The edge is the data pipeline, not the arithmetic.

7 Conclusion

The TU-P/E reframes a familiar multiple around an observable, pre-GAAP signal of margin inflection. The next steps are replacing the simulated universe with the live EDGAR + AAS pipeline, estimating s per sector, and running the quarterly-rebalance evaluation on point-in-time data.
